

Are current temp patterns significantly different from historical?

Objective: Test whether current max temps are significantly higher than historical max temps

Steps:

1. Collect raw max temps measured during field season
2. Pull in raw historical data
3. Trim raw historical data to max/min of field season julian date
4. Run model $\text{temp} \sim \text{julian date (continuous)} + \text{present/historical (discrete)}$
5. Evaluate: if CI of coefficient of present/historical overlaps 0, no significant difference.
6. May need to include interaction $\text{julian date} * \text{present/historical}$, if we think that it's important to consider that the slope of temp on julian date might be different between present and past (maybe we do think this, I'll just include it to be safe.)

Step 1: Collect raw max temps measured during field season Step 2: Pull in raw historical data Step 3: Trim raw historical data to max/min of field season julian date

```
t <- read_rds("../data/temp.rds") %>%
  filter(logger_position == "0") %>%
  dplyr::select(date, maxt) %>%
  mutate(jdate = yday(date),
         era = "present") %>%
  dplyr::select(jdate, maxt, era)

maxjdate <- max(t$jdate, na.rm = TRUE)
minjdate <- min(t$jdate, na.rm = TRUE)

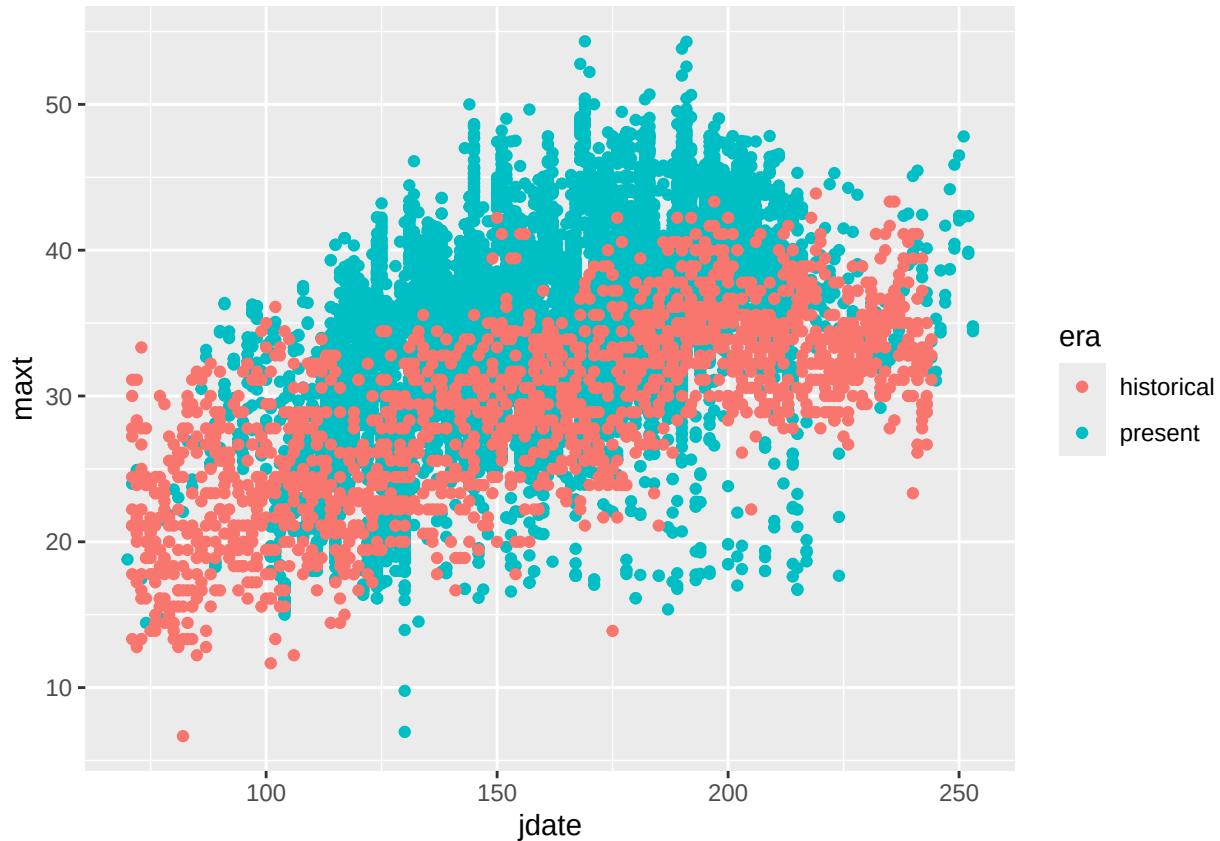
h <- read_rds("../data/cimis_1883_dailymax.rds") %>%
  mutate(jdate = yday(date)) %>%
  filter(jdate > minjdate, jdate < maxjdate) %>%
  dplyr::select(jdate, maxt) %>%
  mutate(era = "historical")

t_full <- t %>% rbind(h)
```

Step 4: Run model $\text{temp} \sim \text{julian date (continuous)} + \text{present/historical (discrete)}$ Step 5: Evaluate: if CI of coefficient of present/historical overlaps 0, no significant difference. Step 6: May need to include interaction $\text{julian date} * \text{present/historical}$, if we think that it's important to consider that the slope of temp on julian date might be different between present and past (maybe we do think this, I'll just include it to be safe.)

```
ggplot(data = t_full,
       mapping = aes(x = jdate, y = maxt, color = era)) +
  geom_point()
```

```
## Warning: Removed 347 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



```
m <- lm(maxt ~ jdate_scaled * era + era, data = t_full %>% mutate(jdate_scaled = scale(jdate)))
summary(m)
```

```
##
## Call:
## lm(formula = maxt ~ jdate_scaled * era + era, data = t_full %>%
##   mutate(jdate_scaled = scale(jdate)))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -24.0327  -3.5944   0.0554   3.9448  19.0011
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    28.80458    0.13292  216.701 < 2e-16 ***
## jdate_scaled     3.01323    0.08739   34.480 < 2e-16 ***
## erapresent      5.09104    0.14222   35.798 < 2e-16 ***
## jdate_scaled:erapresent 0.66085    0.10466    6.314 2.8e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.505 on 13568 degrees of freedom
## (347 observations deleted due to missingness)
## Multiple R-squared:  0.322, Adjusted R-squared:  0.3219
## F-statistic: 2148 on 3 and 13568 DF, p-value: < 2.2e-16
```

2021-2023 temps appear to be elevated over historical temps. However, it's possible that differences in

measurement devices could cause this difference. To check, I'll repeat this analysis with cimis data 2021-2023.

```
t <- read_rds("../data/temp.rds") %>%
  filter(logger_position == "0") %>%
  dplyr::select(date, maxt) %>%
  mutate(jdate = yday(date),
         era = "nestbox") %>%
  dplyr::select(jdate, maxt, era)

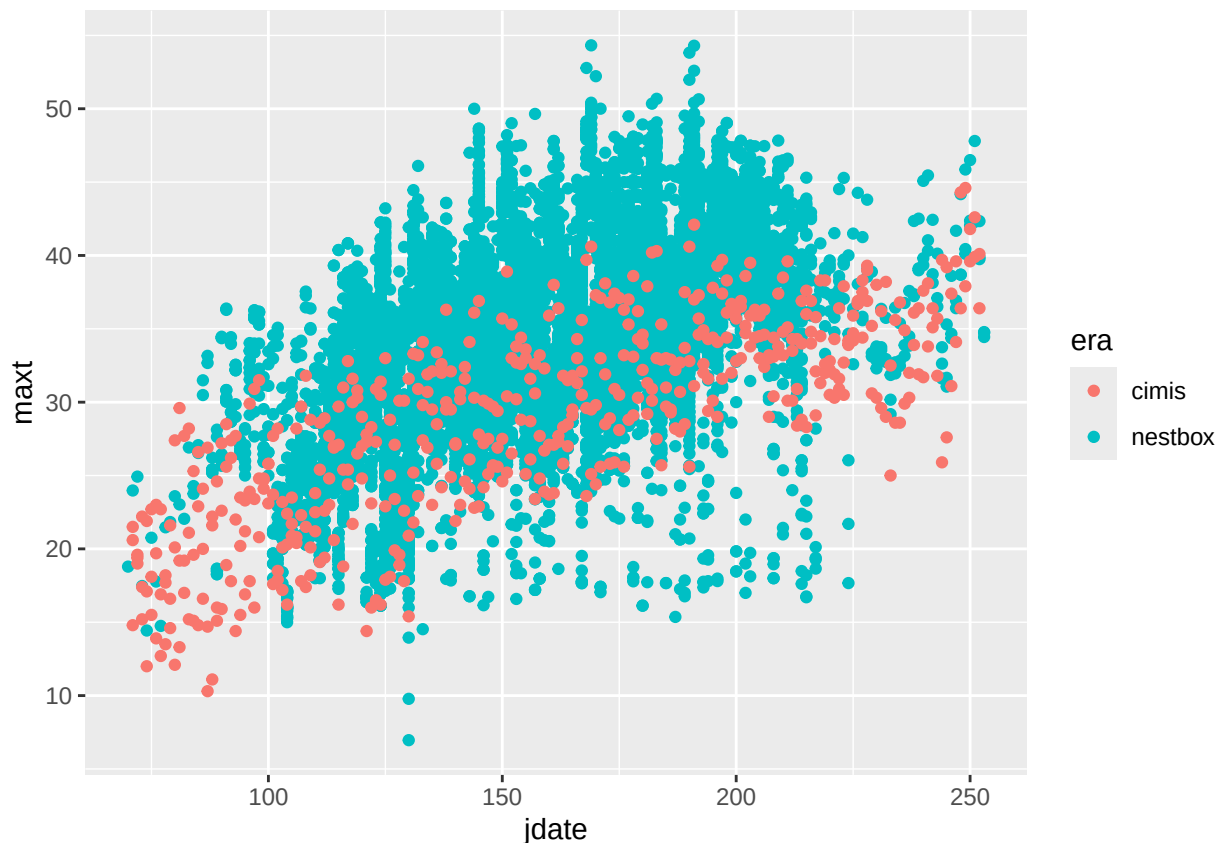
maxjdate <- max(t$jdate, na.rm = TRUE)
minjdate <- min(t$jdate, na.rm = TRUE)

h <- read_csv("../data/Davis_CIMIS_2021-2023.csv") %>%
  dplyr::select(Date, `Max Air Temp (C)`) %>%
  rename(date = Date, maxt = `Max Air Temp (C)`) %>%
  mutate(date = mdy(date)) %>%
  mutate(jdate = yday(date)) %>%
  filter(jdate > minjdate, jdate < maxjdate) %>%
  dplyr::select(jdate, maxt) %>%
  mutate(era = "cimis")

## Rows: 900 Columns: 7
## -- Column specification -----
## Delimiter: ","
## chr (5): Stn Name, CIMIS Region, Date, Jul, qc
## dbl (2): Stn Id, Max Air Temp (C)
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

t_full <- t %>% rbind(h)

ggplot(data = t_full,
       mapping = aes(x = jdate, y = maxt, color = era)) +
  geom_point()
```



```
m <- lm(maxt ~ jdate_scaled * era + era, data = t_full %>% mutate(jdate_scaled = scale(jdate)))
summary(m)
```

```
##
## Call:
## lm(formula = maxt ~ jdate_scaled * era + era, data = t_full %>%
##   mutate(jdate_scaled = scale(jdate)))
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-24.0327	-3.6380	0.0822	4.0210	19.0011

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	28.7062	0.2448	117.274	<2e-16 ***
jdate_scaled	3.0995	0.1459	21.248	<2e-16 ***
eranestbox	5.1852	0.2501	20.732	<2e-16 ***
jdate_scaled:eranestbox	0.2634	0.1554	1.695	0.0901 .

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.589 on 12372 degrees of freedom
## Multiple R-squared:  0.2795, Adjusted R-squared:  0.2793
## F-statistic: 1600 on 3 and 12372 DF,  p-value: < 2.2e-16
```

So in fact the differences between past and present were probe differences entirely!